

Module 3

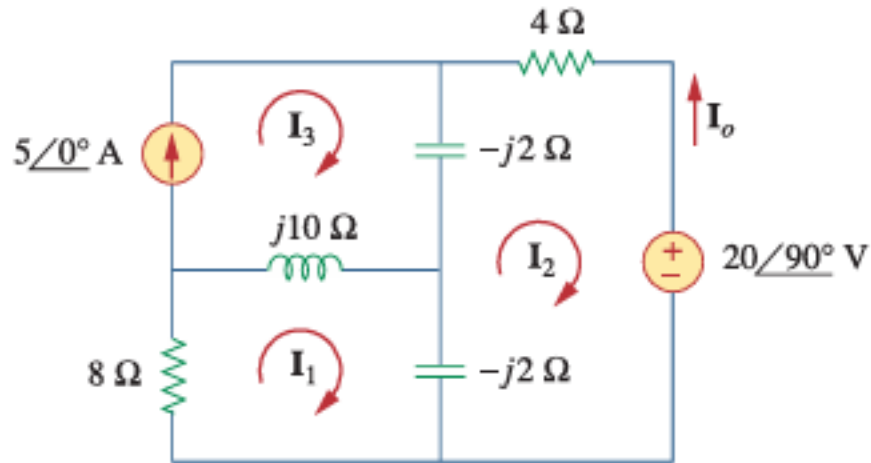
Sinusoidal Steady-State Analysis

Dr. Elizabeth Rufus

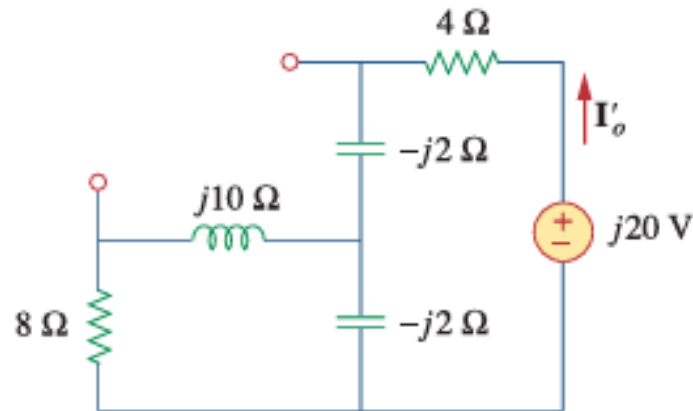
Professor, School of Electronics Engineering

Superposition theorem

1. Use Superposition theorem to find the current I_o .



$$\mathbf{I}_o = \mathbf{I}'_o + \mathbf{I}''_o$$

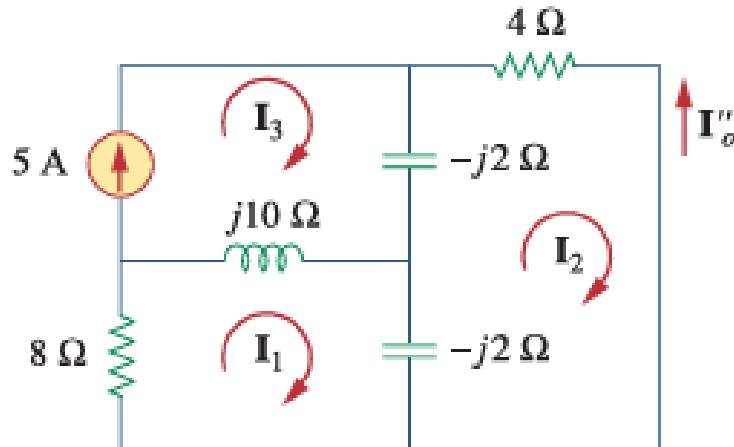


$$\mathbf{Z} = \frac{-j2(8 + j10)}{-2j + 8 + j10} = 0.25 - j2.25$$

$$\mathbf{I}'_o = \frac{j20}{4 - j2 + \mathbf{Z}} = \frac{j20}{4.25 - j4.25}$$

$$\mathbf{I}'_o = -2.353 + j2.353$$

Superposition theorem



From eqn. 2 and eqn.3,

$$(4 - j4)\mathbf{I}_2 + j2\mathbf{I}_1 + j10 = 0$$

$$\mathbf{I}_1 = (2 + j2)\mathbf{I}_2 - 5 \quad \text{---4}$$

substituting eqn. 3 and eqn.4 in eqn.1,

$$(8 + j8)[(2 + j2)\mathbf{I}_2 - 5] - j50 + j2\mathbf{I}_2 = 0$$

$$\mathbf{I}_2 = \frac{90 - j40}{34} = 2.647 - j1.176$$

$$\mathbf{I}''_o = -\mathbf{I}_2 = -2.647 + j1.176$$

$$\mathbf{I}_o = \mathbf{I}'_o + \mathbf{I}''_o = -5 + j3.529 = 6.12 \angle 144.78^\circ \text{ A}$$

For mesh 1

$$(8 + j8)\mathbf{I}_1 - j10\mathbf{I}_3 + j2\mathbf{I}_2 = 0 \quad \text{---1}$$

For mesh 2

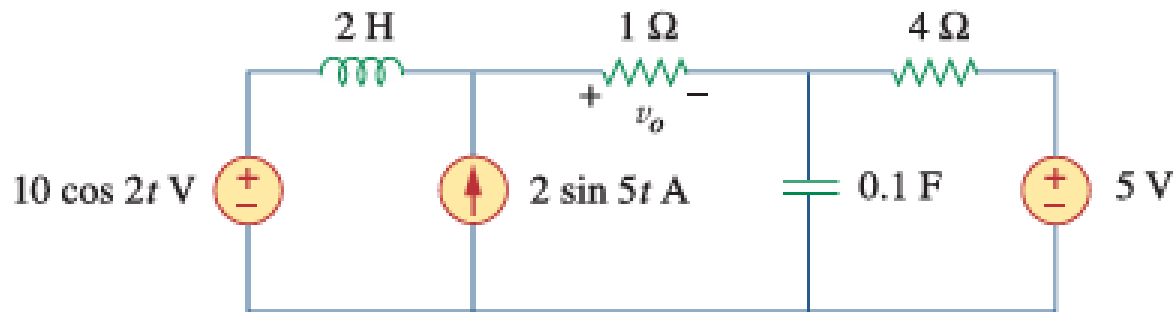
$$(4 - j4)\mathbf{I}_2 + j2\mathbf{I}_1 + j2\mathbf{I}_3 = 0 \quad \text{---2}$$

For mesh 3

$$\mathbf{I}_3 = 5 \quad \text{---3}$$

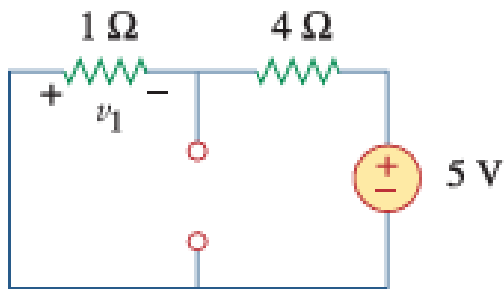
Superposition theorem

2. Find v_o of the circuit using Superposition theorem.



$$v_o = v_1 + v_2 + v_3$$

Note that the circuit operates at 3 different frequencies.

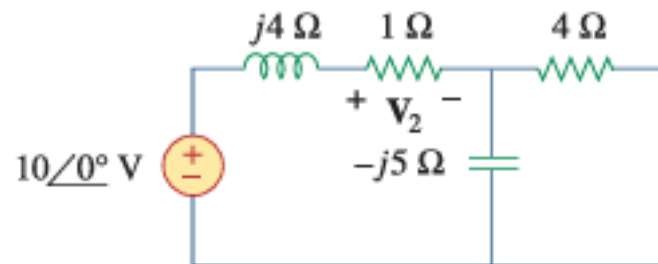


Setting all the sources to 0 and continuing with 5 V source, at steady state the capacitor is open and the inductor is short to DC or **Since $\omega = 0$, $j\omega L = 0$, $1/j\omega C = \infty$**

By voltage division,

$$-v_1 = \frac{1}{1 + 4}(5) = 1 \text{ V}$$

Superposition theorem



$$\begin{aligned}
 10 \cos 2t &\Rightarrow 10\angle 0^\circ, & \omega &= 2 \text{ rad/s} \\
 2 \text{ H} &\Rightarrow j\omega L = j4 \Omega \\
 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j5 \Omega
 \end{aligned}$$

$$\mathbf{Z} = -j5 \parallel 4 = \frac{-j5 \times 4}{4 - j5} = 2.439 - j1.951$$

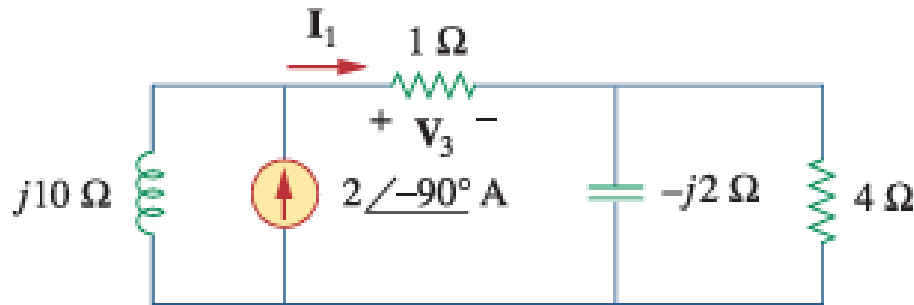
By voltage division,

$$\mathbf{V}_2 = \frac{1}{1 + j4 + \mathbf{Z}} (10\angle 0^\circ) = \frac{10}{3.439 + j2.049} = 2.498\angle -30.79^\circ$$

In time domain,

$$v_2 = 2.498 \cos(2t - 30.79^\circ)$$

Superposition theorem



$$\begin{aligned}
 2 \sin 5t &\Rightarrow 2 \angle -90^\circ, & \omega &= 5 \text{ rad/s} \\
 2 \text{ H} &\Rightarrow j\omega L = j10 \Omega \\
 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j2 \Omega
 \end{aligned}$$

$$\mathbf{Z}_1 = -j2 \parallel 4 = \frac{-j2 \times 4}{4 - j2} = 0.8 - j1.6 \Omega$$

By current division,

$$\begin{aligned}
 \mathbf{I}_1 &= \frac{j10}{j10 + 1 + \mathbf{Z}_1} (2 \angle -90^\circ) \text{ A} \\
 \mathbf{V}_3 &= \mathbf{I}_1 \times 1 = \frac{j10}{1.8 + j8.4} (-j2) = 2.328 \angle -80^\circ \text{ V}
 \end{aligned}$$

In time domain,

$$v_3 = 2.33 \cos(5t - 80^\circ) = 2.33 \sin(5t + 10^\circ) \text{ V}$$

$$v_o(t) = -1 + 2.498 \cos(2t - 30.79^\circ) + 2.33 \sin(5t + 10^\circ) \text{ V}$$

Next class: Source Transformation